

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, PREDICTIVE QUOTIENTS, AND RESOURCE-AWARE REDUCTION

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ABSTRACT. We prove matching finite-state prediction laws for separated Markov repellers with forbidden transitions, bounded conformal distortion, and delay-observable Hölder readouts. Acquisitions may have nongeometric renewal times. A strict renewal-energy inequality makes the entire adaptive admissible tree suffix closed, with linear state count even at unary vertices. Its geometric profile is comparable to weighted-orbit quantization and to optimal causal risk over all randomized time-dependent observers. Two pressure roots determine the squared-error exponent. On Markov similarity systems, polynomial renewal tails produce distinct power-log and iterated-log critical corrections, with a uniform finite-size profile window. A second realization theorem uses posterior contraction rather than a common regeneration component. An inward finite quantizer gives matched $M^{-2/d}$ excess risk on unbounded filtering models, including Gaussian hidden chains with no global minorization at any fixed number of steps. Exact simulations compare finite-memory interfaces.

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1. INTRODUCTION

The information retained by a causal observer must serve predictions whose time of use is not known when that information is acquired. A description of the current state can therefore be inadequate even when it is an optimal quantization at the acquisition time. There is also a

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converse difficulty: a good finite description of an entire future orbit need not admit a closed update on the same finite state set. These two questions link the geometry of prediction to the implementation of a finite memory.

This paper develops two realization principles for that link. The first is an exact suffix-closure principle for renewal-weighted cylinder codes on a mixing Markov repeller. The second is a contracting posterior principle for continual partial observation on an unbounded state space. Both are compared with the full class of randomized, time-dependent observers having a bounded persistent register.

1.1. The intrinsic law. Let Σ_A be a topologically mixing subshift of finite type, with forbidden transitions allowed, and let K be its strongly separated geometric realization by contractions of bounded conformal distortion. The class includes similarities in any Euclidean dimension and nonconstant smooth inverse derivatives in dimension one. The measure on K is a stationary Hölder Gibbs law. A Hölder observable may be noninjective provided a finite delay vector recovers the geometric state with a uniform snowflake lower bound.

Exact acquisitions occur at independent renewal times. The survival law need not be geometric. A finite cylinder energy combines the mass of the acquired cylinder, the successive suffix contraction scales, and the tail of the stationary renewal age. Splitting maximal-energy cylinders defines a finite profile G_M . Our first conclusion is

$$\mathcal{R}_M \asymp e_M^2(Z_w^f) \asymp G_M,$$

where Z_w^f is the weighted observable orbit and $\mathcal{R}_M$ is the optimal long-time average prediction risk. The constants hold for every budget, not merely along a sequence of cylinder depths.

The realization argument must accommodate vertices with a unique successor and suffix cylinders with a unique predecessor. Strict inequality of cylinder masses is then unavailable. We instead prove strict decrease of the renewal distortion under prefix attachment. Together with stationarity, it forces the complete greedy tree to be suffix closed. Unary vertices are counted, and the whole state set has size at most a graph-dependent constant times the number of leaves. The lower bound uses separated tubes in Hilbert space and is valid for arbitrary off-image centres. Thus neither admissibility nor causal execution is hidden in the definition of a static codebook.

When the renewal weights have exponential rate q , the squared-error exponent is determined by the larger of two pressure roots,

$$P_A(s_g\varphi + 2\beta s_g\psi) = 0, \quad P_A(s_t\varphi) + s_t \log q = 0.$$

The first combines mass with geometric contraction; the second combines mass with survival through unobserved dynamics. We prove the existence of the quantization exponent from these equations. It is not an additional hypothesis on an infinite-dimensional orbit image. For a Markov source, zeros of the transition matrix remain zeros in the transfer matrices.

The first-order exponent does not determine the critical correction. For the Parry law on any primitive graph and equal-ratio similarities, renewal survival $q^k(k+1)^{-\kappa}$ gives, at the common critical value, corrections $(\log M)^{1-\kappa}$ for $\kappa < 1$, $\log \log M$ for $\kappa = 1$, and a bounded factor for $\kappa > 1$. The underlying geometry and pressure threshold can be kept fixed. A uniform finite-size profile limit describes the window $q_n = q_c e^{x/n}$. These statements describe actual renewal laws and matching risk orders; they do not assert an optimal leading constant from a pair of inequalities.

1.2. A nonregenerative realization principle. For continual observations, let m_t denote the true complete-history conditional mean. Suppose it obeys a finite-dimensional Borel recursion which contracts conditionally in fourth mean in its previous mean coordinate. Individual updates may expand. An inward rational quantizer preserves fourth-moment bounds on an unbounded state space. The resulting recursive observer attains excess squared risk $O(M^{-2/d})$ using at most M persistent values. If the stationary posterior mean has positive density on a cube, a matching

lower bound holds against all observers. The Bayes floor here is the *full-history* Bayes error, not the error of the latest observation.

For a stationary Gaussian autoregression with Gaussian observation noise, all hypotheses are verified and the risk is $P + \Theta(M^{-2})$, where P is the steady conditional variance. Every fixed-step hidden transition kernel in this example lacks a nonzero global common minorizing measure. The realization proof is therefore genuinely different from inserting an independent reference experiment at each acquisition. The general theorem states its posterior contraction hypothesis explicitly; it is not a contraction theorem for all nonlinear filters.

1.3. The resource and its invariances. Only the number of values of the persistent state is constrained. Fixed programs, fixed real constants, the absolute clock, and transient computation are separate resources. A renewal indicator is an input to the update, not a free post-update observation. Theorem 7.1 proves exact operational simulations: adding a K -valued event to the decoder costs at most a factor K in state count; splitting registers costs their product; binary checkpoint encodings are related by $M = 2^B$. These transformations preserve the exponent and, under fixed-budget dilation, the risk order. A real-valued free observation at decoding is a different experiment, as the explicit uniform example shows.

1.4. Relation to previous work. Thermodynamic formulas for *spatial* quantization predate the present work. Lindsay and Mauldin [11] relate quantization dimension for finite conformal systems to the temperature function. Atnip, Roychowdhury and Urbański [12, Theorem 3.1] extend that relation to an infinite conformal family under their summability and strong open set hypotheses. Roychowdhury [9] treats Gibbs-like measures on cookie-cutter sets. Those spatial results include systems beyond the finite, strongly separated class used here. We do not claim a generalization of their geometric hypotheses. The additional object here is the survival-weighted observable *future orbit*, with an unrestricted causal converse and an admissible, suffix-closed finite-state realization. Classical pressure and greedy variable-to-fixed trees [5, 10] are inputs, not inventions of this paper.

Finite-memory and zero-delay approximation likewise have a substantial literature. Linder and Yüksel [6] and Wood, Linder and Yüksel [7] establish structural and stationary coding results. Ghomi, Linder and Yüksel [13, Theorem 4] prove stationary optimality for linear vector Markov sources and control finite-horizon approximation to the infinite-horizon distortion. Their channel alphabet and their history-dependent coding policy are not a single finite persistent register. Our filtering result is not a new proof of stationary zero-delay coding optimality.

Kara and Yüksel [2, Theorem 12] prove exponentially accurate finite-memory feedback under their POMDP stability hypotheses. Cregg, Alajaji and Yüksel [14, Theorem 2 and Corollary 2] bound the discounted suboptimality of sliding-window source-channel policies by predictor stability, with an exponential window bound under sufficient Dobrushin conditions. Thus stability-based finite-memory approximation is already established. In contrast, Theorem 6.1 concerns the number of retained values, a matched squared-error exponent against the full-history Bayes floor, and an unbounded continuous observation model. It is a quantitative realization result in that resource, not a Q-learning or optimal-control theorem. Recursive filter quantization itself is also classical; see Pagès and Pham [1].

Static functional quantization [8], dynamical rate distortion [4], and data-rate control [3] constrain different objects. The distinction is operational: a block code or information-rate bound does not by itself specify a closed update on a bounded persistent state set. Conversely, the state-count theorems here give no runtime, program-description, or channel-capacity bound.

1.5. Organization and dependencies. Sections 2–4 prove the Markov profile and pressure theorems. Section 5 gives the critical classification and uniform window. Sections 6 and 7 prove the nonregenerative and interface results. The general posterior-orbit converse is included with its complete proof in Section 8. The complete edition additionally retains the earlier Gibbs,

dependent-acquisition, noisy expansion, calibration, root-collision, filter, control and foundational developments. They have their own hypotheses; the current Markov and filtering proofs do not depend on the singular-statistics or unbounded-operator conclusions.

The programme's T07–T08 question is the passage from acquired predictive geometry to actual finite-state evolution. The two realization mechanisms proved here resolve that question in their stated classes, including a noncompact nonregenerative model. The historical physical models require further, model-specific results. The active A2 conductor and nonreduced-generator edition [15] is an algebraic input to a different observation interface: its scheme multiplicities do not specify a noisy observation kernel. The accompanying snapshot audit records that interface at its current source edition rather than identifying algebraic multiplicity with a statistical exponent. The complete edition also proves a joint sample–label deficiency bound for its actual known-mark count-to-contact protocol. The overflow alphabet and the comparison randomization are explicitly charged and distinguished from the physical statistic.

2. MARKOV GEOMETRY AND THE FINITE-STATE LAW

We first specify the geometry independently of the observation apparatus. Let A be a primitive zero–one matrix on an alphabet $\mathcal{A}$ of cardinality $b \geq 2$, with Perron root $\lambda_A > 1$. Write Σ_A for its one-sided shift and $\mathcal{L}_n$ for its words of length n . The empty word is denoted by $\emptyset$. Forbidden transitions are permitted; in particular, a vertex may have only one successor.

Let $D \subset \mathbb{R}^d$ be compact, of diameter at most one, and let $h_i : D \rightarrow D$ be injective contractions. Assume that the compact sets $h_i(D)$ have pairwise distance at least $g > 0$. For $v = i_0 \cdots i_{n-1}$ write $h_v = h_{i_0} \circ \cdots \circ h_{i_{n-1}}$. We require scales $r_\emptyset = 1$, $0 < r_- \leq r_i \leq r_+ < 1$, and $D_0 \geq 1$ such that

$$D_0^{-1} r_v |x - y| \leq |h_v x - h_v y| \leq r_v |x - y|, \quad (2.1)$$

$$D_0^{-1} r_u r_v \leq r_{uv} \leq r_u r_v. \quad (2.2)$$

These conditions are imposed for words for which the compositions are used. They hold for similarities in every dimension, and for increasing one-dimensional $C^{1+\gamma}$ contractions with uniformly positive derivative and bounded logarithmic distortion, with $r_v = \sup |h'_v|$. For higher-dimensional conformal maps, (2.1) is an explicit metric bounded-distortion hypothesis; no global co-Lipschitz conclusion is inferred merely from a pointwise derivative estimate.

The coding map $\pi : \Sigma_A \rightarrow K \subset D$ is injective. Set $T\pi(\omega) = \pi(\sigma\omega)$ and $K_v = \pi([v])$. Let μ be the image of a stationary, fully supported Hölder Gibbs measure on Σ_A . Normalize its potential φ by $P_A(\varphi) = 0$. In particular, for a fixed C_G ,

$$C_G^{-1} e^{S_n \varphi(\omega)} \leq \mu[v] \leq C_G e^{S_n \varphi(\omega)} \quad (\omega \in [v]). \quad (2.3)$$

We use the classical Gibbs and variational pressure theorems for a primitive finite shift in the form recorded in [5]. The profile theorem below needs only stationarity and a uniform positive lower bound on each admissible one-letter conditional probability.

2.1. Observables and acquisition times. Fix $0 < \beta \leq 1$, put $\tau = 2\beta$, and let $f : K \rightarrow \mathbb{R}^m$ satisfy

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad (2.4)$$

$$\sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_o |x - y|^{2\beta} \quad (x, y \in K) \quad (2.5)$$

for some finite J and $c_o > 0$. Thus the present output need not identify the state; finitely many successive outputs must do so quantitatively. Condition (2.5) concerns the actual observable on K , not an assigned distance on symbolic words.

At a renewal an independent state U of law μ is observed exactly. Between renewals the state evolves by T . Cycle lengths are independent copies of a positive integer L , independent of the acquired states, with finite mean. The age weights and their tails are

$$w_k = \frac{\mathbb{P}(L > k)}{\mathbb{E}L}, \quad W(n) = \sum_{k \geq n} w_k, \quad 0 < \underline{q} \leq \frac{w_{k+1}}{w_k} \leq \bar{q} < 1. \quad (2.6)$$

An initial acquisition occurs at zero; the next renewal time is not announced. Nongeometric lengths satisfying (2.6) are allowed. Write X_t for this process.

An observer keeps I_t in a set of at most M elements. Its update may use I_{t-1} , absolute time, the renewal indicator, and the acquired state at a renewal. The post-update decoder uses I_t , absolute time, and independent coding randomness, but does not separately receive the renewal indicator or a retained real-valued observation. All observationally dependent persistent variables are counted in I_t . Stochastic kernels, private randomness, arbitrary fixed real constants, and transient computation are allowed. Section 7 compares this convention with other finite-memory interfaces.

Set

$$\begin{aligned} Z_w^f(x) &= (\sqrt{w_k} f(T^k x))_{k \geq 0} \in \ell^2(\mathbb{N}_0; \mathbb{R}^m), \\ e_M^2(Z_w^f) &= \inf_{c_1, \dots, c_M} \int \min_{i \leq M} \|Z_w^f(x) - c_i\|^2 d\mu(x), \\ \mathcal{R}_M &= \inf_c \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t < N} \mathbb{E} |f(X_t) - \hat{f}_t|^2. \end{aligned} \quad (2.7)$$

Centres in the quantization problem may lie anywhere in the Hilbert space. The infimum in the last line ranges over the entire preceding observer class, not just stationary or cylinder observers.

2.2. A computable geometric profile. For $v \in \mathcal{L}_n$ and its suffix $v_{k:} = i_k \cdots i_{n-1}$ define

$$V_w(v) = \sum_{k < n} w_k r_{v_{k:}}^\tau + W(n), \quad a_w(v) = \mu[v] V_w(v), \quad a_w(\emptyset) = 1. \quad (2.8)$$

Starting from the root, split a leaf of maximal a_w into all its admissible one-symbol successors, with a fixed rule for ties. Let $\mathcal{T}_N$ be the last tree in this sequence having at most N leaves, let $\mathcal{P}_N$ be its leaves, and put

$$G_N(w) = \sum_{v \in \mathcal{P}_N} a_w(v). \quad (2.9)$$

For the finitely many budgets preceding the first root split, the tree is the root. Unary splits are genuine operations. They do not increase the number of leaves, but they cannot occur in an infinite uninterrupted chain. Consequently the last tree in this definition exists.

The construction is a finite mathematical algorithm on cylinder masses, scales and renewal tails. Effective numerical evaluation additionally requires certified access to those real inputs; no computability assertion for arbitrary Borel or Hölder data is implicit.

Theorem 2.1 (Markov renewal profile). *Under (2.1)–(2.6), there are positive constants c, C , independent of M , such that*

$$cG_M(w) \leq e_M^2(Z_w^f) \leq \mathcal{R}_M \leq CG_M(w) \quad (M \geq 1). \quad (2.10)$$

Every randomized time-dependent observer satisfies the middle lower bound with its lower limiting average loss. A stationary deterministic observer attains the upper order.

More precisely, the complete admissible tree $\mathcal{T}_N$ is suffix closed. If it has L leaves, all its vertices can be retained as states, and their number is at most

$$(b+1)(2L-1). \quad (2.11)$$

Its exact number is $1 + \sum_v \text{internal} d^+(v)$, where $d^+(v)$ is the number of children actually present. No time-depth factor is required. The constants in (2.10) may depend on the graph, distortion, separation, conditional masses, observable, and the two ratios in (2.6).

For the pressure statement assume additionally that a Hölder potential ψ satisfies $\log r_v = S_n \psi(\omega) + O(1)$ uniformly on $[v]$, and that

$$\lim_{n \rightarrow \infty} n^{-1} \log w_n = \log q < 0. \quad (2.12)$$

The inverse-derivative potential gives ψ in the one-dimensional smooth case. For similarities, $\psi(\omega) = \log r_{\omega_0}$.

Theorem 2.2 (Intrinsic pressure exponent). *There are unique roots $s_g, s_t \in (0, 1)$ of*

$$P_A(s_g \varphi + \tau s_g \psi) = 0, \quad P_A(s_t \varphi) + s_t \log q = 0. \quad (2.13)$$

Writing $s_* = \max(s_g, s_t)$, one has

$$\lim_{M \rightarrow \infty} \frac{-\log \mathcal{R}_M}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z_w^f)}{\log M} = \frac{1 - s_*}{s_*}. \quad (2.14)$$

The spatial and survival roots exchange dominance at $q_c = \exp\{-P_A(s_g \varphi)/s_g\}$. This formula uses the exponential rate of the renewal tail. Polynomial factors in that tail can change the critical correction without changing either root.

3. RENEWAL ENERGIES AND ADMISSIBLE SUFFIX TREES

The main combinatorial fact is a strict energy inequality, not strict inequality of cylinder masses. The distinction matters at vertices having a unique predecessor.

Lemma 3.1 (Strict suffix domination). *If u is nonempty and uv is admissible, then*

$$V_w(v) - V_w(uv) \geq (1 - r_+^\tau) r_v^\tau \sum_{k < |u|} w_k > 0, \quad a_w(uv) < a_w(v). \quad (3.1)$$

There are constants $a_0 > 0$ and $\rho < 1$ such that each admissible child satisfies

$$a_0 a_w(v) \leq a_w(vi) \leq \rho a_w(v), \quad \sum_{i: vi \text{ admissible}} a_w(vi) \leq \rho a_w(v). \quad (3.2)$$

The same inequalities hold at the root after decreasing a_0 .

Proof. Write $m = |u|$, $n = |v|$ and suppress the subscript w . Every suffix of v has scale at least r_v . Since the weights decrease,

$$\begin{aligned} V(v) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \left(\sum_{j < n} (w_j - w_{m+j}) + W(n) - W(m+n) \right) \\ &= r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

The remaining prefix terms in $V(uv)$ are at most $r_v^\tau \sum_{k < m} w_k r_{u_k}^\tau$, which is at most $r_+^\tau r_v^\tau \sum_{k < m} w_k$. This proves the first inequality, also for $v = \emptyset$. Stationarity gives $\mu[uv] \leq \mu\{\omega : \omega_{m:m+n} = v\} = \mu[v]$. Both masses are positive, so the second inequality follows. Neither argument completes an inadmissible word by an artificial transition.

For $|v| = n$, appending i multiplies each old spatial scale by a factor between $D_0^{-1} r_-$ and r_+ . The old tail $W(n)$ is replaced by $w_n r_i^\tau + W(n+1)$. The weight assumptions imply

$$(1 - \bar{q})W(n) \leq w_n, \quad W(n+1) \geq \underline{q}W(n).$$

It follows that

$$\ell_0 V(v) \leq V(vi) \leq \rho V(v), \quad \ell_0 = \min(D_0^{-\tau} r_-^\tau, \underline{q}), \quad \rho = 1 - (1 - r_+^\tau)(1 - \bar{q}) < 1.$$

A uniform lower bound $\zeta > 0$ on admissible conditional cylinder masses now gives $a_0 = \zeta \ell_0$ in (3.2). The last assertion follows by summing the masses of the children. Such a ζ follows directly from (2.3) and the bounded potential, for admissible extensions only. $\square$

Lemma 3.2 (Balanced trees and controlled dilation). *Let $L_N = |\mathcal{P}_N|$. For all sufficiently large N , $N/b \leq L_N \leq N$. If θ_N is the energy of the last split vertex, then every leaf satisfies*

$$a_0 \theta_N \leq a_w(v) \leq \theta_N. \quad (3.3)$$

For every fixed $B \geq 1$, with constants allowed to depend on B ,

$$G_{\lceil BN \rceil}(w) \asymp_B G_N(w). \quad (3.4)$$

The complete tree with L leaves has at most $(b+1)(2L-1)$ vertices.

Proof. Maximal split energies never increase, by (3.2). A present leaf was born with energy at least a_0 times an earlier split energy, which proves the lower bound in (3.3); the upper bound follows from the final maximal split. Each split adds between zero and $b-1$ leaves. The split following $\mathcal{T}_N$ increases the number beyond N ; hence $L_N > N - b + 1$, which implies the stated coarser bound.

Here are the finiteness details needed in the presence of unary vertices. A path of b successive vertices all having outdegree one would repeat a vertex and enter a closed deterministic cycle. This contradicts primitivity and $\lambda_A > 1$. Thus a unary chain has length at most b . After suppressing unary vertices, any finite tree with L leaves and no internal vertex of degree one has at most $2L-1$ vertices. Reinserting the unary chains, including an initial chain, gives the asserted bound. In particular an infinite sequence of splits with bounded leaf number is impossible.

Choose ℓ such that A^ℓ has all entries positive. From every last symbol there are at least two distinct continuations of length ℓ . Thus every vertex has at least 2^j descendants at depth ℓj . In continuing a greedy tree with threshold θ_N , if all later leaves have energy less than $a_0^{\ell j+1} \theta_N$, then every original leaf must have been split through those ℓj levels. This would give at least $2^j L_N$ later leaves. Choosing j so that $2^j > b(B+2)$ shows that the final maximum leaf energy at budget $\lceil BN \rceil$ is at least a fixed multiple of θ_N . The balance estimate at that budget bounds its minimum leaf energy by another fixed multiple of θ_N . Since $L_{\lceil BN \rceil}$ is at least a fixed multiple of N , its energy sum is at least $c_B N \theta_N \geq c_B G_N$. Conversely splitting decreases the sum by (3.2), so $G_{\lceil BN \rceil} \leq G_N$. Increasing constants handles the finitely many initial trees. $\square$

Lemma 3.3 (Observable orbit cylinders). *The image $Z_w^f(K_v)$ has squared diameter at most $CV_w(v)$. If two different sequences have longest common prefix u , then*

$$\|Z_w^f(x) - Z_w^f(y)\|^2 \geq cV_w(u). \quad (3.5)$$

All constants depend only on the displayed geometric and observable hypotheses, not on the word.

Proof. Introduce the auxiliary snowflake orbit distance $d_\beta(x, y)^2 = \sum_{k \geq 0} w_k |T^k x - T^k y|^{2\beta}$. The Hölder estimate bounds $\|Z_w^f(x) - Z_w^f(y)\|^2$ from above by $H^2 d_\beta(x, y)^2$. Multiplying (2.5) at $T^k x, T^k y$ by w_k and summing gives

$$c_o d_\beta(x, y)^2 \leq \sum_{j=0}^J \sum_{k \geq 0} w_k |f(T^{k+j} x) - f(T^{k+j} y)|^2 \leq \left(\sum_{j=0}^J \underline{q}^{-j} \right) \|Z_w^f(x) - Z_w^f(y)\|^2.$$

For points in K_v , the first n terms of d_β^2 are bounded by $w_k r_{v_k}^{2\beta}$ and its remaining terms by $W(n)$. If u is their longest common prefix, then at time $|u|$ they lie in different first-level images, and

at time $k < |u|$ their distance is at least $D_0^{-1}gr_{u_k}$. The first post-prefix term is $w_{|u|}g^{2\beta}$, and $W(|u|) \leq w_{|u|}/(1 - \bar{q})$. These estimates prove both assertions, including the empty prefix. $\square$

Lemma 3.4 (Unrestricted quantization and suffix realization). *The quantization errors satisfy $e_M^2(Z_w^f) \asymp G_M(w)$. Every complete greedy tree is suffix closed, including its leaves. A tree with L leaves gives a deterministic observer with at most $(b+1)(2L-1)$ states and average loss at most $C \sum_{v \in \mathcal{P}} a_w(v)$.*

Proof. Taking one orbit representative from each leaf cylinder proves $e_N^2 \leq CG_N$. For the reverse direction use the leaf partition $\mathcal{P}_{4bM}$, which has at least $4M$ members, apart from finitely many budgets absorbable in the constants. Around $Z_w^f(K_v)$ take its open Hilbert tube of radius $\delta\sqrt{V_w(v)}$. These tubes are disjoint for a sufficiently small fixed δ : for incomparable leaves the separation in (3.5) is at their common prefix, and appending symbols only decreases V_w . Any M Hilbert centres, whether or not on the orbit image, occupy at most M tubes. On every other cylinder the nearest-centre squared error is at least $\delta^2 V_w(v)$. The balance estimate shows that these unoccupied cylinders carry at least a fixed fraction of the energy sum. Hence $e_M^2 \geq cG_{4bM} \geq c'G_M$ by (3.4). For each of the finitely many omitted budgets this error is positive: the delay condition and infinitely supported Gibbs law exclude support on finitely many orbit points.

Suppose an internal word uv , with u nonempty, is about to be split. If its suffix v has not yet been split, some current leaf t is a prefix of v . By (3.2) and (3.1), $a_w(t) \geq a_w(v) > a_w(uv)$, contradicting maximality. Thus every proper suffix of an internal word is already internal. If $v = i_0 \cdots i_{n-1}$ is a leaf with $n \geq 2$, the word $i_1 \cdots i_{n-2}$ is a suffix of its internal parent, so it has been split. Its admissible child $i_1 \cdots i_{n-1}$ is therefore present. For $n = 1$ the suffix is the root. Iteration proves closure of the entire tree; it does not require that all symbols follow all others.

Choose $x_v \in K_v$ for every nonempty tree vertex, and decode that vertex as $f(x_v)$. At an acquisition store the unique leaf prefix of the observed sequence. At each nonacquisition delete the first symbol. Decode the empty word as any fixed $f(x_*)$, and keep it until the next acquisition. Each update stays within the same tree. If the acquired word is v of length n , the squared loss at age $k < n$ is at most $H^2 r_{v_k}^{2\beta}$; after exhaustion it is bounded by a constant. The renewal-reward formula therefore bounds the average by $C \sum_v \mu[v]V_w(v)$. All vertices are charged, with the count established in Lemma 3.2. $\square$

Proof of Theorem 2.1. The retained posterior-orbit theorem, Theorem 8.2, applies with exact observation and Bayes floor zero. It gives $e_M^2 \leq \mathcal{R}_M$ and the stated lower-limit assertion for arbitrary stochastic kernels. Its full proof is included in both reading editions. Lemma 3.4 gives the other lower bound. Choose a leaf budget N a fixed sufficiently small fraction of M so that (2.11) is at most M . The same lemma and fixed-budget dilation give an upper bound $CG_N \leq C'G_M$. A one-state observer handles the remaining bounded range of M . This proves (2.10). $\square$

4. TWO PRESSURES ON AN ADMISSIBLE LANGUAGE

All pressures in this section are taken on Σ_A . A Hölder potential F has uniformly bounded oscillation of $S_n F$ on cylinders, and its partition sums are comparable to $e^{nPA(F)}$, uniformly in n . These are classical finite-state thermodynamic facts [5]. The new issue is the renewal-weighted sum of suffix scales, which is not an additive potential.

Lemma 4.1 (Mixed partition pressure). *For $0 < s < 1$, under (2.12),*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{v \in \mathcal{L}_n} a_w(v)^s = \max\{P_A(s\varphi + \tau s\psi), P_A(s\varphi) + s \log q\}. \quad (4.1)$$

Proof. Put $P_g(s) = P_A(s\varphi + \tau s\psi)$ and $P_0(s) = P_A(s\varphi)$. The first and last terms in (2.8), separately, imply

$$\sum_{|v|=n} a_w(v)^s \geq c_s \max\{w_0^s e^{nP_g(s)}, W(n)^s e^{nP_0(s)}\}.$$

For the upper bound use $(\sum x_k)^s \leq \sum x_k^s$ and split an admissible word after k letters. Its mass and suffix scale are, within fixed factors, the exponential of the corresponding Birkhoff sums. Dropping the constraint joining the two pieces can only increase the sum. Thus

$$\sum_{|v|=n} a_w(v)^s \leq C_s \left(\sum_{k < n} w_k^s e^{kP_0(s)} e^{(n-k)P_g(s)} + W(n)^s e^{nP_0(s)} \right). \quad (4.2)$$

No factorization of arbitrary words across a forbidden transition is asserted. Given $\delta > 0$, the weight assumption gives $w_k \leq C_\delta e^{k(\log q + \delta)}$, and $W(n) \asymp w_n$. The upper sum has at most $n + 1$ terms with exponential rate at most $\max(P_g(s), P_0(s) + s \log q + s\delta)$. The lower bound has rate at least the same maximum with $\delta = 0$. Let $\delta \downarrow 0$. $\square$

Proof of Theorem 2.2. We first verify uniqueness of the roots. Every vertex has at least two continuations of a common length ℓ . By the positive conditional mass bound, no particular continuation of that length has conditional probability larger than $1 - \zeta^\ell$. Hence $\mu[v] \leq C\eta^{|v|}$ for an $\eta < 1$. The Gibbs estimate gives $S_n\varphi \leq n \log \eta + O(1)$ on every cylinder. Similarly $S_n\psi \leq n \log r_+ + O(1)$. The pressure functions in (2.13) are continuous and strictly decreasing, with positive value $\log \lambda_A$ at zero. At one their values are strictly negative, since $P_A(\varphi) = 0$. Their roots lie in $(0, 1)$.

For $s > s_*$, both exponents in (4.2) are negative. Choose $q'' > q$ close enough that they remain negative with q'' in place of q . Bounding w_k and $W(k)$ by a constant times $(q'')^k$ and summing (4.2) over all n yields

$$\sum_{v \in \mathcal{L}} a_w(v)^s < \infty. \quad (4.3)$$

For a balanced partition with $L \asymp N$ leaves, this implies $L(a_0\theta_N)^s \leq C_s$, and therefore $G_N \leq L\theta_N \leq C'_s N^{-(1-s)/s}$.

For the lower bound at s_g , use the Gibbs probability ν_g for the zero-pressure potential $s_g\varphi + \tau s_g\psi$. On each cylinder,

$$\nu_g[v] \leq C\mu[v]^{s_g} r_v^{\tau s_g} \leq C' a_w(v)^{s_g}.$$

Summing over $\mathcal{P}_N$ gives $1 \leq C' L\theta_N^{s_g}$. Balance then gives $G_N \geq cN^{-(1-s_g)/s_g}$.

For the survival root, a polynomial correction to w_n prevents using q^n as a uniform lower bound. Instead fix $0 < q' < q$. There is $c_{q'} > 0$ with $W(n) \geq c_{q'}(q')^n$ for all n . Let $s_t(q')$ be the zero of $P_A(s\varphi) + s \log q'$, and use the Gibbs measure of this zero-pressure potential. Its cylinder masses are bounded by $C\mu[v]^{s_t(q')}(q')^{|v|s_t(q')}$, hence by $C' a_w(v)^{s_t(q')}$. The same balanced-partition argument proves the lower bound with exponent $(1 - s_t(q'))/s_t(q')$. The roots vary continuously with q' , by strict monotonicity. Letting $q' \uparrow q$ and $s \downarrow s_*$ proves the common exponent for G_N . Theorem 2.1 transfers it to both errors. Finally $s_t = s_g$ precisely when $P_A(s_g\varphi) + s_g \log q = 0$, proving the threshold formula. $\square$

Corollary 4.2 (Markov matrices with zero entries). *Suppose the h_i are similarities and μ is a stationary Markov measure with transition probabilities $p_{ij} > 0$ exactly when $A_{ij} = 1$. The roots in (2.13) are characterized by*

$$\text{spr}((A_{ij} p_{ij}^{s_g} r_i^{\tau s_g})_{ij}) = 1, \quad q^{s_t} \text{spr}((A_{ij} p_{ij}^{s_t})_{ij}) = 1. \quad (4.4)$$

Only admissible entries are powered; other entries are zero. At $s = 0$ the limiting matrix is A , not a matrix of ones.

Proof. Take $\varphi(\omega) = \log p_{\omega_0\omega_1}$ on admissible edges and $\psi(\omega) = \log r_{\omega_0}$. The nonuniform stationary initial factor and the terminal factor are bounded above and below, so do not change pressure. The finite transfer matrices of these edge potentials are exactly those in (4.4). Perron–Frobenius theory and Theorem 2.2 give the assertion. $\square$

Example 4.3 (A noninjective planar readout with a forbidden transition). On $[0, 1]^2$ take

$$h_0(x, y) = (ry, rx), \quad h_1(x, y) = (ry + 2/3, rx), \quad r = 1/3, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Rescaling the ambient domain makes its diameter one without changing the argument. Both maps are similarities and their images have a positive horizontal gap. Give the golden-mean shift any stationary Hölder Gibbs law. For $f(x, y) = x$, the relation $f(T(x, y)) = y/r$ on K gives

$$|f(x, y) - f(x', y')|^2 + |f(T(x, y)) - f(T(x', y'))|^2 = (x - x')^2 + r^{-2}(y - y')^2.$$

Thus (2.5) holds with $J = 1$ and $\beta = 1$. The sequences $000\cdots$ and $01000\cdots$ are admissible, have the same first coordinate, and have different second coordinates. Hence f is not injective. This is a higher-dimensional geometric system with a genuinely forbidden transition and a delayed, rather than instantaneously identifying, observable.

5. CRITICAL CORRECTIONS AND A UNIFORM RENEWAL WINDOW

The exponential rate of the acquisition tail does not determine the critical second order. We give a classification in a family allowing arbitrary primitive transition graphs, every ambient dimension in the similarity construction, and delay-observable outputs.

Assume that all branch ratios equal r , and that μ is the Parry measure of Σ_A . Write $\lambda = \lambda_A$ and $a = r^{2\beta}$. For $q \in (0, 1)$ and $\kappa \geq 0$ take the actual survival law

$$\mathbb{P}(L > k) = q^k(k+1)^{-\kappa}, \quad c_{q,\kappa} = \left(\sum_{k \geq 0} q^k(k+1)^{-\kappa} \right)^{-1}. \quad (5.1)$$

Its weights satisfy (2.6), with lower ratio $q2^{-\kappa}$ and upper ratio q . The common depth- n energy scale is

$$V_{q,\kappa}(n) = c_{q,\kappa} \left[a^n \sum_{k < n} (q/a)^k (k+1)^{-\kappa} + \sum_{k \geq n} q^k (k+1)^{-\kappa} \right]. \quad (5.2)$$

Theorem 5.1 (Critical power–log classification). *Let $n_M = \lfloor \log M / \log \lambda \rfloor$. In this family,*

$$\mathcal{R}_M \asymp V_{q,\kappa}(n_M). \quad (5.3)$$

For fixed κ the constants can be chosen uniformly for q in a compact subinterval of $(0, 1)$. The critical value is $q_c = a$. With $\alpha = \log(1/a)/\log \lambda$, at this value, as $M \rightarrow \infty$,

$$\mathcal{R}_M \asymp M^{-\alpha} \begin{cases} (\log M)^{1-\kappa}, & 0 \leq \kappa < 1, \\ \log \log M, & \kappa = 1, \\ 1, & \kappa > 1. \end{cases} \quad (5.4)$$

In particular the same graph, geometry, observable, pressure roots, and critical exponential survival rate admit different critical corrections. The specification of the renewal law is essential.

Proof. Perron eigenvectors give, uniformly in $v \in \mathcal{L}_n$, $\mu[v] \asymp \lambda^{-n}$ and $|\mathcal{L}_n| \asymp \lambda^n$. All these cylinders have scale (5.2). A fixed-depth complete tree has $O(\lambda^n)$ vertices (sum the geometric level counts), is suffix closed, and gives average loss at most $CV_{q,\kappa}(n)$ by the pointwise construction in Lemma 3.4. Choose $n = n_M - C_0$ so that its state count is at most M .

For a lower bound choose $n = n_M + C_1$ with at least $4M$ cylinders. The disjoint tubes used in Lemma 3.4 then leave at least three quarters of the cylinders unoccupied by any M centres. Their masses are comparable and their scales equal, giving $e_M^2 \geq cV_{q,\kappa}(n)$. A fixed change in depth changes V by factors bounded above and below by (3.2)'s scale proof, uniformly for q in the specified compact set. The converse gives (5.3). This argument neither treats the Parry measure as independent nor fills forbidden transitions.

At $q = a$, the first term in (5.2) is $c_{a,\kappa} a^n \sum_{j=1}^n j^{-\kappa}$; the tail is $O(a^n(n+1)^{-\kappa})$. Integral comparison gives the three possibilities in (5.4). Also $P_A(s\varphi) = (1-s)\log\lambda$ and $P_A(s\varphi + 2\beta s\log r) = (1-s)\log\lambda + s\log a$. Thus the two roots coincide at $q = a$, as asserted. $\square$

Theorem 5.2 (Uniform finite-size profile window). *Fix $\kappa \geq 0$ and put $q_n = ae^{x/n}$. Uniformly for x in a bounded real interval, the following limits hold:*

$$\frac{V_{q_n,\kappa}(n)}{a^n n^{1-\kappa}} \longrightarrow c_{a,\kappa} \int_0^1 t^{-\kappa} e^{xt} dt, \quad 0 \leq \kappa < 1, \quad (5.5)$$

$$\frac{V_{q_n,1}(n)}{a^n \log n} \longrightarrow c_{a,1}, \quad \kappa = 1, \quad (5.6)$$

$$\frac{V_{q_n,\kappa}(n)}{a^n} \longrightarrow c_{a,\kappa} \sum_{j \geq 1} j^{-\kappa}, \quad \kappa > 1. \quad (5.7)$$

The optimal risk is uniformly comparable to these profiles at budgets of order λ^n . The displayed leading coefficients are profile coefficients, not asserted exact optimal-risk coefficients.

Proof. The normalizing constants tend uniformly to $c_{a,\kappa}$ by a common geometric majorant. For $0 \leq \kappa < 1$, split the normalized sum at $k = \delta n$. Away from zero, its summands form a uniformly convergent Riemann sum for $t^{-\kappa} e^{xt}$. The part below δn is bounded uniformly by $C\delta^{1-\kappa}$; then let $\delta \downarrow 0$. The normalized tail is $O(n^{-1})$ uniformly, using $q_n \leq q_+ < 1$ and $q_n^n/a^n = e^x$. For $\kappa = 1$,

$$\left| \sum_{k < n} \frac{e^{xk/n} - 1}{k+1} \right| \leq C$$

uniformly on bounded x , while the unperturbed harmonic sum is $\log n + O(1)$. The tail divided by $a^n \log n$ vanishes. For $\kappa > 1$, dominated convergence applies to the summable sequence $(k+1)^{-\kappa}$, and the tail divided by a^n is $O(n^{-\kappa})$. Finally use the uniform comparison in Theorem 5.1. $\square$

The classification concerns the stated Markov–Parry similarity family and the renewal tails (5.1). For unequal Hölder derivative cocycles the finite profile theorem and pressure exponent still apply; (5.4) is not imposed on that larger class. In the geometric case $\kappa = 0$ the first window profile is $(1-a)(e^x - 1)/x$, continuously extended at $x = 0$.

6. FINITE REGISTERS WITHOUT A COMMON REGENERATION LAW

A different realization mechanism applies when observations arrive continually and the posterior itself contracts. It does not use an independent reference component of a transition kernel.

Let $(U_t, Y_t)_{t \in \mathbb{Z}}$ be stationary, with $U_t \in \mathbb{R}^d$ square integrable and Y_t standard Borel. Write

$$\mathcal{Y}_t^- = \sigma(Y_s : s \leq t), \quad m_t = \mathbb{E}[U_t \mid \mathcal{Y}_t^-], \quad B = \mathbb{E}\|U_0 - m_0\|^2.$$

An observer starting at zero receives Y_t at every update and retains at most M states. Its decoder receives only its post-update state, absolute time, and independent coding randomness. Let $\mathcal{R}_M^{\text{fil}}$ be the infimum of its upper limiting average squared loss. The reference B uses the full two-sided past; initial lack of that past will be handled by contraction, not by a fictitious initial sufficient statistic.

Theorem 6.1 (Contracting posterior realization). *Suppose $m_0 \in L^4$ and a Borel function F has a measurable Lipschitz coefficient $\ell(y) \geq 0$ such that*

$$\begin{aligned} m_t &= F(m_{t-1}, Y_t), & \|F(x, y) - F(x', y)\| &\leq \ell(y)\|x - x'\|, \\ \mathbb{E}[\ell(Y_t)^4 \mid \mathcal{Y}_{t-1}^-] &\leq \rho^4 < 1, & K_4 &:= \|F(0, Y_0)\|_{L^4} < \infty. \end{aligned} \quad (6.1)$$

A uniform bound $\ell(y) \leq \rho$ is sufficient but not necessary; individual observation updates may expand. If the law of m_0 has a Lebesgue density bounded below by a positive constant on a cube of positive side length, then

$$B + cM^{-2/d} \leq \mathcal{R}_M^{\text{fil}} \leq B + CM^{-2/d} \quad (M \geq 1). \quad (6.2)$$

The lower bound holds for the lower limiting average loss of every randomized time-dependent observer. The upper observer is stationary and deterministic. It uses no compact support or minorization hypothesis on either the hidden dynamics or the observation process.

For a product register of $(2n - 1)^d$ states, $n \geq 2$, its excess error has the explicit upper bound

$$\limsup_{t \rightarrow \infty} \mathbb{E}\|m_t - \hat{U}_t\|^2 \leq \frac{d(1 + K_4/(1 - \rho))^4}{n^2(1 - \rho)^2}. \quad (6.3)$$

Lemma 6.2 (A finite inward quantizer). *For $n \geq 2$ define*

$$j_n(x) = \min\left(n - 1, \left\lfloor \frac{n|x|}{1 + |x|} \right\rfloor\right), \quad Q_n(x) = \text{sgn}(x) \frac{j_n(x)}{n - j_n(x)}.$$

It has $2n - 1$ values and satisfies

$$|Q_n(x)| \leq |x|, \quad |x - Q_n(x)| \leq n^{-1}(1 + |x|)^2. \quad (6.4)$$

Its coordinatewise extension to $\mathbb{R}^d$ satisfies $\|Q_n x\| \leq \|x\|$ and $\|x - Q_n x\| \leq \sqrt{d} n^{-1}(1 + \|x\|)^2$.

Proof. For $x \geq 0$ put $u = x/(1 + x)$, so the inverse is $g(u) = u/(1 - u)$. The quantized argument $j_n(x)/n$ is at most u and differs from it by less than $1/n$, including the saturated last bin. The increasing derivative $g'(u) = (1 - u)^{-2}$ bounds the difference by $(1 + x)^2/n$. Odd reflection proves the scalar assertion. Sum the squared coordinate errors and use $|x_i| \leq \|x\|$ for the vector bound. $\square$

Proof of Theorem 6.1. Conditional orthogonality, also after adjoining randomness independent of the experiment, gives

$$\mathbb{E}\|U_t - \hat{U}_t\|^2 = B + \mathbb{E}\|m_t - \hat{U}_t\|^2. \quad (6.5)$$

At any fixed t , freeze all independent randomization. The decoder has at most M possible values, even though the encoder has seen a long observation history. On a cube where the density of m_t is at least c_0 , take balls of radius $\delta M^{-1/d}$ around those values. A sufficiently small δ , independent of their locations, makes their total intersection volume at most half the cube's volume. The remaining volume incurs squared distance at least $\delta^2 M^{-2/d}$. Integrating the density lower bound and then the randomization gives the lower bound at each time, and hence for lower limiting averages.

For the upper bound start from $z_{-1} = 0$ and iterate

$$x_t = F(z_{t-1}, Y_t), \quad z_t = Q_n x_t, \quad \hat{U}_t = z_t. \quad (6.6)$$

Only the index of z_t is stored; x_t is transient. The values of Q_n are fixed rational vectors, part of the fixed decoder table. Conditional fourth-moment contraction gives $\|\ell(Y_t)z_{t-1}\|_{L^4} \leq \rho\|z_{t-1}\|_{L^4}$. The inward property and Minkowski's inequality therefore give, without independence of successive observations,

$$\|z_t\|_{L^4} \leq \rho\|z_{t-1}\|_{L^4} + K_4 \leq A_4 := K_4/(1 - \rho), \quad \|x_t\|_{L^4} \leq A_4.$$

Lemma 6.2 therefore implies $\|x_t - Q_n x_t\|_{L^2} \leq \sqrt{d}(1 + A_4)^2/n$. Couple the exact stationary recursion to the same observations. The previous error is measurable with respect to $\mathcal{Y}_{t-1}^-$; conditional Cauchy–Schwarz implies $\mathbb{E}[\ell(Y_t)^2 \mid \mathcal{Y}_{t-1}^-] \leq \rho^2$. Thus

$$\|m_t - z_t\|_{L^2} \leq \rho \|m_{t-1} - z_{t-1}\|_{L^2} + \sqrt{d}(1 + A_4)^2/n.$$

The finite initial error is multiplied by ρ^{t+1} . Summing this recursion proves (6.3). Equation (6.5) gives the upper average bound. For large M choose $n = \lfloor (M^{1/d} + 1)/2 \rfloor$, so $(2n - 1)^d \leq M$ and $n \asymp M^{1/d}$. For the remaining finite range, the one-state output zero has finite loss, and an enlarged constant completes (6.2). $\square$

Corollary 6.3 (An unbounded hidden Gaussian chain). *Let*

$$X_t = aX_{t-1} + \xi_t, \quad Y_t = X_t + \eta_t, \quad 0 < |a| < 1, \quad \xi_t \sim N(0, v), \quad \eta_t \sim N(0, \omega), \quad (6.7)$$

where $v, \omega > 0$, the noises are independent, and X has its stationary law. Let P be the unique positive solution of

$$P = \frac{\omega(a^2 P + v)}{a^2 P + v + \omega}. \quad (6.8)$$

Then $\mathcal{R}_M^{\text{fil}} = P + \Theta(M^{-2})$. For d independent such coordinates the law is $\sum_i P_i + \Theta(M^{-2/d})$. For every fixed $k \geq 1$, the hidden k -step kernel admits no nonzero finite measure dominated by all its transition probabilities.

Proof. Put $S = v/(1 - a^2)$. Conditional Gaussian projection gives the one-step prediction variance $a^2 P + v$ and the update (6.8). The map on its right sends $[0, S]$ into itself and has derivative at most $a^2 < 1$; its unique fixed point lies in $(0, S)$. Iterating from a finite past and letting that past recede gives the stationary conditional variance P and the mean recursion

$$m_t = (1 - K)am_{t-1} + KY_t, \quad K = \frac{a^2 P + v}{a^2 P + v + \omega}.$$

This passage is also the L^2 martingale convergence of the increasing observation sigma fields. Gaussian conditioning supplies the displayed finite-past recursion at each step. Thus no distributional filter approximation is being substituted for the true conditional mean. The mean is Gaussian with variance $S - P > 0$, so has all required moments and a strictly positive density on every bounded interval. The contraction constant is $|(1 - K)a| < 1$. Theorem 6.1 applies; independent coordinates give the vector statement with the Euclidean contraction constant equal to the largest coordinate constant.

The hidden k -step transition from x is a Gaussian with mean $a^k x$ and fixed finite positive variance. If a nonzero finite measure ν were dominated by every such probability, some bounded Borel set E would have $\nu(E) > 0$. Sending x to infinity gives $Q^k(x, E) \rightarrow 0$, a contradiction. The proof of the risk law therefore cannot be an application of a global one-step or fixed-multistep common-reference minorization. $\square$

The contraction and Gaussian conditioning in this section are classical mechanisms. The quantitative assertion is the matched error in the number of values of the *whole persistent register*, with a uniform-in-time implementation on an unbounded state space. Constants in (6.3) display their deterioration as $\rho \uparrow 1$. No assertion at $\rho = 1$, nor a reinterpretation of this theorem as the $\varepsilon \downarrow 0$ limit of a minorization bound, is needed.

7. OPERATIONAL CHANGES OF THE FINITE-MEMORY INTERFACE

The following comparison is distribution-free. In particular it applies to both realization mechanisms above.

Theorem 7.1 (Finite interface equivalence). *Consider any sequential experiment with a standard Borel update input. Suppose an event E_t in an alphabet of size $K < \infty$ is available to the update. Let R_M^{side} be the optimum when the post-update decoder also receives E_t , and let R_M^{reg} be the optimum when it does not. With the same choices of time and independent randomness, for every nonnegative path loss and every horizon, and likewise for upper or lower limiting average loss,*

$$R_{KM}^{\text{reg}} \leq R_M^{\text{side}} \leq R_M^{\text{reg}}. \quad (7.1)$$

These inequalities arise from exact pathwise simulations. An M -value register has an equivalent binary checkpoint encoding in $\lceil \log_2 M \rceil$ bits with unused strings allowed. Several finite registers of sizes $M_1, \dots, M_s$ are equivalent to one register of size $\prod_i M_i$.

Proof. An observer can ignore a free event, giving the second inequality. For the first, store the pair (I_t, E_t) . Its transition discards the previous event, performs the original update using the current input, and stores the new event. The original decoder is then evaluated on this pair. The outputs agree for each input and each realization of all coding random variables. Thus all path losses and their expectations agree, not just a selected one-time distortion. Applying a fixed bijection to the finite state set transports the transition kernels and decoder to binary strings or to a product state set. The inverse bijections give the remaining assertions. $\square$

Consequently a positive squared-error exponent is invariant under any fixed finite event interface, fixed regrouping of finite registers, and finite-state renaming. When the risk or its Bayes excess satisfies a fixed-budget-dilation comparison, its full order is invariant as well. For a binary checkpoint budget B the corresponding register scale is $M = 2^B$; this changes a polynomial in M to an exponential in B , rather than changing the theorem's physical resource silently.

Example 7.2 (An unbounded side interface is a different experiment). Let X_t be independent uniform random variables on $[0, 1]$ and let the current update input be $Y_t = X_t$. A decoder with the real-valued Y_t as a free post-update side input has zero loss with one persistent state. Without that side input the optimal M -state loss is $1/(12M^2)$: midpoint quantization attains it, and for cells of masses p_i the uniform interval moment inequality gives $\sum_i p_i^3/12 \geq 1/(12M^2)$. Randomization does not improve the pointwise nearest-centre minimum. A finite K in (7.1) cannot be replaced by an arbitrary standard Borel alphabet. The distinction is the presence of a new measurement at the decoding boundary, not a hidden age-state convention.

8. POSTERIOR ORBITS AND FINITE-STATE REALIZATION

Let E and $\mathcal{Y}$ be standard Borel spaces, let μ be a probability on E , and let K be a probability kernel from E to $\mathcal{Y}$. Let $T : E \rightarrow E$ and $f : E \rightarrow \mathbb{R}^r$ be Borel, with $\|f\| \leq F < \infty$. At each renewal a fresh pair (U, Y) has law $\mu(du)K(u, dy)$. The observer acquires Y , not necessarily U . Between renewals the hidden state follows T . The successive cycle lengths $L_0, L_1, \dots$ are independent copies of a positive integer-valued random variable L , independent of all acquired pairs, and

$$\ell = \mathbb{E}L < \infty, \quad s_k = \mathbb{P}(L > k), \quad w_k = \frac{s_k}{\ell}, \quad W(n) = \sum_{k \geq n} w_k. \quad (8.1)$$

Thus $\sum_{k \geq 0} w_k = 1$. There is an acquisition at time 0. Subsequent acquisition times are $\tau_j = L_0 + \dots + L_{j-1}$. At time $t = \tau_j + k < \tau_{j+1}$, the target is $f(T^k U_j)$. The renewal indicator is observed, but the length of the current cycle is not announced at its beginning.

An M -state observer retains $I_t \in \{1, \dots, M\}$. Its update is an arbitrary Borel stochastic kernel of the preceding state, absolute time, renewal indicator, and, at a renewal only, the new observation. After the update, the decoder has the state, absolute time, and independent public randomness. It does not separately retain the last observation, the renewal indicator, or the renewal age. Private randomness is allowed, but any data-dependent persistent register, including

an age or a random-seed position, is part of I_t . Program descriptions, real constants, and transient computation are not charged to M . This is a persistent-cardinality constraint, not a bound on total computational storage.

Choose simultaneous Borel versions

$$m_k(y) = \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0} \in \mathcal{H} := \ell^2(\mathbb{N}_0; \mathbb{R}^r). \quad (8.2)$$

A countable union of null sets makes these versions simultaneous; values there may be set to zero. Write

$$e_M^2(Z) = \inf_{c_1, \dots, c_M \in \mathcal{H}} \mathbb{E} \min_{i \leq M} \|Z(Y) - c_i\|_{\mathcal{H}}^2, \quad B = \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \quad (8.3)$$

The number B is the unavoidable error of the complete acquisition history. Let $A_t = t - \max\{\tau_j : \tau_j \leq t\}$, and let Y_t^* be the observation at that last renewal. Define

$$E_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|m_{A_t}(Y_t^*) - \hat{f}_t\|^2, \quad (8.4)$$

$$R_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|f(X_t) - \hat{f}_t\|^2. \quad (8.5)$$

Here and below an infimum is over the specified machine class; it need not be attained. All statements about e_M^2 use squared distortion.

Lemma 8.1 (Kernel realization). *The preceding machine class can be realized using a countable family of independent uniform random variables, independent of the experiment. After fixing this family and the history before a renewal, the first n outputs along a continuation with no further renewal belong to a set of at most M vectors in $(\mathbb{R}^r)^n$.*

Proof. For a finite-state transition kernel the inverse cumulative probabilities give a Borel function of the input and one uniform variable. For a general standard Borel output, embed its range as a Borel subset of $[0, 1]$, use the measurable inverse distribution function, and map back; a fixed default value handles the null exceptional set. Applying this construction to each time-indexed kernel uses only countably many uniforms. The product probability space supplies them jointly. Public uniforms are included in the same construction, with their prescribed sharing between update and decoder. Once they are fixed, the update and decoder are deterministic. Given the state immediately after a renewal, the no-renewal input sequence and absolute times determine the subsequent states and outputs. There are only M possible starting states. The argument uses a fixed time-indexed family, not an uncharged data-dependent position in a random tape. $\square$

Theorem 8.2 (Posterior-orbit converse and realization). *For the experiment just defined,*

$$R_M^{\text{av}} = B + E_M^{\text{av}}, \quad e_M^2(Z) \leq E_M^{\text{av}}. \quad (8.6)$$

In fact every randomized, time-dependent M -state machine satisfies the latter lower bound with its $\liminf$ time-average excess loss. Conversely, for all integers $N, n \geq 1$,

$$E_{Nn+1}^{\text{av}} \leq e_N^2(Z^{<n}) + F^2 W(n) \leq e_N^2(Z) + F^2 W(n), \quad (8.7)$$

where $Z^{<n}$ is the first n coordinates of Z . The upper bound is realized, up to an arbitrary positive error, by a stationary deterministic machine with states $\{1, \dots, N\} \times \{0, \dots, n-1\} \cup \{\partial\}$. In particular, for $M \geq 2$,

$$e_M^2(Z) \leq E_M^{\text{av}} \leq \inf_{1 \leq n \leq M-1} \left\{ e_{\lfloor (M-1)/n \rfloor}^2(Z) + F^2 W(n) \right\}. \quad (8.8)$$

Proof. Conditional on the full acquired history, the renewal indicators, and independent coding randomness, the conditional mean of $f(X_t)$ is $m_{A_t}(Y_t^*)$. Indeed, the current acquired pair is independent of previous cycles, and survival of its cycle is independent of that pair. Conditional orthogonality gives, for every machine,

$$\mathbb{E}\|f(X_t) - \hat{f}_t\|^2 = \mathbb{E}\|f(X_t) - m_{A_t}(Y_t^*)\|^2 + \mathbb{E}\|m_{A_t}(Y_t^*) - \hat{f}_t\|^2.$$

The Cesàro renewal-reward identity makes the average of the first term tend to B . Bounded rewards and $\mathbb{E}L < \infty$ suffice, including arithmetic or deterministic cycle lengths. This proves the equality in (8.6).

Fix n . At a renewal at time s , condition on the past and then use Lemma 8.1. Along the first n no-renewal steps the outputs are chosen from at most M decoder strings. The new Y has its original marginal law under this conditioning. Since survival to age k has probability s_k , the expected excess reward in those steps is at least

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_i \sum_{k < n} s_k \|m_k(Y) - c_{i,k}\|^2 = \ell e_M^2(Z^{<n}).$$

Randomizing the selection of a string cannot reduce its pointwise minimum. Integrating the conditioned variables preserves the bound. Let $u_s = \mathbb{P}(s \text{ is a renewal})$. Rewards arising from distinct last renewals are disjoint. Summing over $0 \leq s \leq N - n$ therefore gives a lower bound $\ell e_M^2(Z^{<n}) \sum_{s=0}^{N-n} u_s$ for the total excess loss up to time $N - 1$. The elementary renewal theorem, in its Cesàro form, gives $N^{-1} \sum_{s < N} u_s \rightarrow 1/\ell$. Consequently the lower limit of average loss is at least $e_M^2(Z^{<n})$.

Truncation cannot increase quantization error. Conversely, extending a codebook for the truncation by zero gives

$$e_M^2(Z^{<n}) \leq e_M^2(Z) \leq e_M^2(Z^{<n}) + F^2 W(n).$$

Letting $n \rightarrow \infty$ proves the converse, without a compactness assumption on the orbit image or a restriction to on-image centres.

For the upper bound choose an ε -optimal N -centre codebook for $Z^{<n}$. Coordinatewise projection permits its k th coordinate to be written $\sqrt{w_k} c_{i,k}$ with $\|c_{i,k}\| \leq F$; a zero-weight coordinate is immaterial. At a renewal, choose the first nearest centre, a Borel choice, and set the state to $(i, 0)$. At a nonrenewal change (i, k) to $(i, k + 1)$ until $k = n - 1$, after which the state is ∂ . The latter state is absorbing between renewals. Decode (i, k) as $c_{i,k}$ and ∂ as zero. The average excess reward is at most $e_N^2(Z^{<n}) + \varepsilon + F^2 W(n)$ by the same renewal-reward identity. This is an actual finite-state update; its age information is precisely the charged second coordinate. Let $\varepsilon \downarrow 0$. Unused states are permitted, which proves the last assertion. $\square$

Corollary 8.3 (Invariance of the quantization exponent). *Suppose $W(n) \leq A \exp(-cn^\beta)$ for some $A, c, \beta > 0$, and*

$$\alpha = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z)}{\log M} \quad \text{exists in } (0, \infty). \quad (8.9)$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log E_M^{\text{av}}}{\log M} = \alpha. \quad (8.10)$$

If $e_M^2(Z) \asymp M^{-\alpha}$, more explicitly

$$c_1 M^{-\alpha} \leq E_M^{\text{av}} \leq C_1 M^{-\alpha} (\log M)^{\alpha/\beta} \quad (M \geq 2),$$

with constants depending on the displayed hypotheses. For geometric renewals with parameter $p > 0$, the exponent conclusion also holds for the infimum of the supremum-in-time excess risk.

Proof. Choose any $s > \alpha$ and $n = \lceil ((s/c) \log M)^{1/\beta} \rceil$. Then $\log[(M-1)/n]/\log M \rightarrow 1$, whereas $W(n) \leq AM^{-s}$. Apply (8.8); the two logarithmic bounds converge to α . Under the two-sided power bound the same substitution gives the explicit inequalities, enlarging constants for finitely many small M .

For geometric renewals put $q = 1 - p$, so $w_k = pq^k$. A stationary reset-based machine has a nonnegative age-loss sequence d_k . With an initial acquisition at zero its loss at time t is

$$p \sum_{k < t} q^k d_k + q^t d_t \leq p^{-1} \sum_{k \geq 0} w_k d_k.$$

Use the upper machines in Theorem 8.2; the converse follows because a supremum is at least the lower time-average. $\square$

The corollary is an exponent theorem, not a uniform equivalence of errors for arbitrary orbits. It applies to arbitrary Borel dynamics and arbitrary noisy or partial reset observations. Its geometric invariant is the quantization exponent of the *posterior* future orbit; the tail assumption controls the cost of making a finite codebook causal. Computing that invariant in a particular dynamical system remains a separate mathematical task.

Proposition 8.4 (Exact autonomous realization). *Let a ray mean a sequence $c = (c_k)_{k \geq 0}$ in the ball $\{z \in \mathbb{R}^r : \|z\| \leq F\}$. Among stationary deterministic machines, the optimal average excess risk is*

$$\inf_{\substack{|C| \leq M \\ SC \subseteq C}} \mathbb{E} \min_{c \in C} \sum_{k \geq 0} w_k \|m_k(Y) - c_k\|^2, \quad (Sc)_k = c_{k+1}. \quad (8.11)$$

The reset update may depend on the previous state; allowing that dependence does not alter this infimum.

Proof. For a fixed autonomous no-renewal update a and decoder d , a state i determines the ray $(d(a^k i))_{k \geq 0}$. These at most M rays are closed under S . Any reset rule selects one of them, possibly using the previous state; its reward is no smaller than the pointwise minimum over all the rays. Conversely, for a finite shift-closed set, retain the chosen ray as the state, apply S on a nonrenewal, and decode its first coordinate. The first nearest ray is a Borel reset rule independent of the previous state. Renewal reward proves the formula. This is a characterization of the stated autonomous class, not an assertion that all optimal randomized nonstationary machines are autonomous. $\square$

Example 8.5 (Orbit cardinality does not include an age register). Let $E = \{0, 1\}$, $U = 1$ deterministically, $T(0) = T(1) = 0$, and $f(x) = x$. Use geometric renewals with $0 < p < 1$ and a constant reset observation. The posterior orbit is the single ray $(1, 0, 0, \dots)$, so $e_1^2(Z) = 0$ and $B = 0$. Nevertheless a one-state decoder, having only absolute time and independent public coins, does not know whether a renewal occurred at that time. For every $t \geq 1$ the minimum squared error is $p(1 - p)$, since the target is an independent Bernoulli(p) renewal indicator. Two states suffice for zero error: write 1 at each renewal and 0 otherwise. Thus the orbit converse cannot be reversed with the same cardinality without a realizability condition. The finite-chain overhead in (8.7) and the closure in (8.11) address a genuine obstruction.

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